

Construction Wage Cyclicalality

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ECON 490: Final Capstone Presentation

Introduction

Construction is one of the most cyclical labor markets in the U.S.

Public belief: wages rise in housing booms and fall in busts.

Research Question:

Do male construction workers earn a wage premium, and does that premium vary with housing market cycles?

Background

Why study construction worker wages?

*Construction is one of the most cyclical industries in the U.S.
Demand rises and falls with the housing market, which is notoriously volatile.*

What the public conversation gets wrong... *and how can I add to it?*

*Policy debates often frame construction labor issues around pay levels
("Are construction workers underpaid?")*

But in a cyclical sector, hourly wages may be stable while hours and job security collapse when markets weaken.

LA considers new minimum wage for city's housing construction workers

By
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Topline:

Should construction workers on mid-sized Los Angeles apartment buildings earn at least \$32.35 per hour? The L.A. City Council is scheduled to vote Tuesday on whether to order a study of the proposed new industry-specific minimum wage.

Background

Why hourly wage premiums matter

Hourly wages reflect how the market values construction skills, independent of hours worked.

If no wage premium exists—even during strong housing periods—it signals that cyclical risk isn't being compensated.

Key idea motivating this project

*If construction is risky work, a wage premium **should** appear.*

*If it **doesn't**, that has major implications for worker welfare and policy.*

Data EXTERNAL (IPUMS CPS 2008-2022)

Outcome Variable - Log hourly wage

Focuses on men ages 25 - 64 to reflect the majority composition of construction labor.

Key Explanatory Variables - Construction Worker (construction_worker)

*Binary Indicator = 1 if employed in industry code 770
(Construction)*

Control Variables

Age & Age², Full-time status, Education level, State fixed effects, Year fixed effects

These ensure that wage comparisons are not simply driven by demographics, education differences, or long-term trends.

Data (ZHVI₂₀₀₈₋₂₀₂₂)

Housing Market Conditions (measure of cyclicality)

Derived from Zillow ZHVI data (state × year)

Good Market = state-year where home prices grew > 2% YoY

Interaction:

Construction Worker × Good Market

Tests whether construction wages respond differently to housing cycles.

Data Structure

Each row = one male worker in the CPS, merged with his state-year's housing market indicators.

Final sample sizes:

All workers: 687,032

Blue-collar comparison group: 95,570

Methods

First Stage: Baseline Wage Premium

Goal: compare hourly wages between construction workers and similar male workers

Controls for: age, education, and full-time status, and includes state and year fixed effects

Answers the question: “Is there a construction wage premium?”

Second Stage: Role of Housing Cycles

Goal: link each worker to their state-year housing market conditions and test whether the wage premium changes in stronger markets.

Adds an interaction between: construction work and housing-market growth

This stage answers: “Does the premium expand or shrink when housing markets boom?”

Methods - *note on Blue-Collar Comparison Sample

**I repeat both stages using a restricted sample of blue-collar, goods-producing industries.*

This allows me to see whether construction wages differ from similar labor-intensive jobs, rather than from all workers.

First Stage Results: Baseline Construction Wage Premium

Goal: Estimate whether male construction workers earn more or less than similar non-construction workers.

	All Workers (M1)	Blue Collar (M1)
Construction Worker	-0.015 (0.005)	-0.026 (0.023)
	(0.010)	(0.273)
Num.Obs.	687032	95570

These findings suggest the construction wage penalty is small but negative.

Next step: look into how this penalty changes across housing cycles

Across both comparison groups, construction workers **do not** earn a wage premium; if anything, they earn **slightly lower** hourly wages than similar workers.

Second Stage Results

	All Workers (M3)	Blue Collar (M3)
Construction Worker (CW)	-0.021	-0.018
Good Market	0.002	-0.007
CW × Good Market	0.005	-0.015
Num.Obs.	651370	90420

Key findings

Estimated wage premium/penalty stays small and mostly negative, even in good housing markets.

The Good Market and CW × Good Market coefficients are not statistically different from zero (insignificant)

Interpretation

Housing booms don't really raise hourly pay for male construction workers

Benefits show up more in hours worked and employment, not in the wage rate

Why my Results Differ from my Outline

- Early outline used limited samples and no fixed effects, making construction appear to earn a premium.
- Full model uses:
 - state + year fixed effects
 - proper controls (including the decision to study men only)
 - much larger sample

Result: Premium basically disappears or becomes slightly negative, indicating initial outline overstated earnings due to omitted variable bias.

Limitations/ Sources of Bias

1. Cash-heavy labor market not captured

CPS misses under-the-table pay, overtime, per-diem, reimbursements.

Real earnings for some construction workers may be higher than reported

2. Hourly wage $\neq$ annual income

Construction is highly seasonal; hours fluctuate widely.

A worker may have high hourly pay but low annual earnings.

3. State-level housing data is too broad

Housing cycles vary within a state (e.g., L.A. vs Bakersfield).

4. Construction treated as one group

Mixes high-skill trades (electricians) and low-skill laborers. more; ignoring this biases results.

5. Unobserved migrant labor supply shocks

Construction booms often attract temporary/migrant labor → increases supply, pushes wages down, masks boom-time wage growth.

6. Housing-cycle measurement

ZHVI YoY growth is simple... better would be multi-year cycles or local permitting data

Conclusion

Research Question:

Do male construction workers earn a wage premium, and does that premium vary with housing market cycles?

*Construction workers **do not** earn a meaningful wage premium - hourly wages are slightly **lower** after accounting for age, education, and full-time status.*

*Housing booms **do not** raise construction wages; pay remains stable even in strong markets, suggesting labor supply adjusts instead of wages.*